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Time Table

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MAIR12-CT-1 Max. Marks: 20 Date: 14 October 2026 Time: 11:30–12:30

1. Let $W = \operatorname{span}\{(1,1,0,1),(2,1,1,3),(3,2,1,4)\} \subset \mathbb{R}^4$. Find a basis and the dimension of W. If $W^\perp = \{x \in \mathbb{R}^4 : \langle x, w\rangle = 0 \text{ for every } w \in W\}$, find a basis of $W^\perp$ and verify that $\dim W + \dim W^\perp = 4$.

2. For $u = (u_1, u_2), v = (v_1, v_2) \in \mathbb{R}^2$, define $\langle u, v\rangle_\alpha = u_1v_1 + \alpha(u_1v_2 + u_2v_1) + u_2v_2$. Find all α for which this is an inner product. For $\alpha = \frac{1}{2}$, apply Gram–Schmidt to $u_1 = (1,0), u_2 = (1,1)$ and obtain an orthonormal basis.

3. Define $T : \mathbb{R}^3 \to P_2$ by $T(a,b,c) = (a+b) + (b+c)x + (a+2b+c)x^2$. Find bases and dimensions of $\ker T$ and $\operatorname{Range}(T)$. Hence, using rank–nullity, determine whether T is one-to-one and onto.

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